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Inventor(s)

Kaya; Cetin et al.

METHODS AND APPARATUS TO REGULATE TRANSISTOR SWITCHING

Abstract

An example apparatus includes: current source circuitry having a control terminal; charging circuitry having a first terminal and a second terminal; capacitor circuitry having a first terminal, a second terminal, a third terminal, and a fourth terminal, the first terminal of the capacitor circuitry coupled to the second terminal of the charging circuitry, the second terminal of the capacitor circuitry coupled to the third terminal of the charging circuitry; and comparator circuitry having a first terminal, a second terminal, and an output terminal, the first terminal of the comparator circuitry coupled to the third terminal of the capacitor circuitry, the second terminal of the comparator circuitry coupled to the fourth terminal of the capacitor circuitry, the output terminal of the comparator circuitry coupled to the control terminal of the current source circuitry.

Inventors: Kaya; Cetin (McKinney, TX), Nitish; FNU (Dallas, TX), Chen; Yang-Lin (Taoyuan City, TW)

Applicant: Texas Instruments Incorporated (Dallas, TX)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This patent application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/559,475 filed Feb. 29, 2024, which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This description relates generally to switching and, more particularly, to methods and apparatus to regulate transistor switching.

BACKGROUND

[0003] As electronics continue to advance, systems have become capable of safely operating at increasingly complex operating conditions, such as higher powers and higher speeds. In driver circuitry, increasingly complex circuitry implements advanced techniques for regulating a supply of power to a load. Such circuitry allows the driver circuitry to precisely control and regulate transistor switching despite complex operating conditions.

SUMMARY

[0004] For methods and apparatus to regulate transistor switching, an example apparatus includes current source circuitry having a control terminal; charging circuitry having a first terminal and a second terminal; capacitor circuitry having a first terminal, a second terminal, a third terminal, and a fourth terminal, the first terminal of the capacitor circuitry coupled to the second terminal of the charging circuitry, the second terminal of the capacitor circuitry coupled to the third terminal of the charging circuitry; and comparator circuitry having a first terminal, a second terminal, and an output terminal, the first terminal of the comparator circuitry coupled to the third terminal of the capacitor circuitry, the second terminal of the comparator circuitry coupled to the fourth terminal of the capacitor circuitry, the output terminal of the comparator circuitry coupled to the control terminal of the current source circuitry. Other examples are described.

[0005] For methods and apparatus to regulate transistor switching, an example apparatus includes an capacitor circuitry including: a set of first capacitors; and a set of second capacitors; charging circuitry coupled to the capacitor circuitry, the charging circuitry configured to: generate a reference voltage by discharging the set of first capacitors; and generate a switching voltage by charging the set of second capacitors; and comparator circuitry coupled to the capacitor circuitry, the comparator circuitry configured to decrease a drive current responsive to a comparison of the reference voltage to the switching voltage. Other examples are described.

[0006] For methods and apparatus to regulate transistor switching, an example apparatus includes a transistor having a control terminal; and current source circuitry having a first terminal and a control terminal, the first terminal of the current source circuitry coupled to the control terminal of the transistor; comparator circuitry having a first terminal, a second terminal, and an output terminal, the output terminal of the comparator circuitry coupled to the control terminal of the current source circuitry; and capacitor circuitry having a first terminal and a second terminal, the first terminal of the capacitor circuitry coupled to the first terminal of the comparator circuitry, the second terminal of the capacitor circuitry coupled to the second terminal of the comparator circuitry. Other examples are described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a block diagram of an example drive system including gate driver circuitry that uses example current multiplication circuitry and an auxiliary current source to adjust a drive current to reduce stress on a motor during switching.

[0008] FIG. 2 is a schematic diagram of the example drive system of FIG. 1.

[0009] FIG. 3 is a schematic diagram of a first configuration of the example drive system of FIGS. 1 and 2.

[0010] FIG. 4 is a schematic diagram of a second configuration of the example drive system of FIGS. 1, 2, and 3.

[0011] FIG. 5 is a schematic diagram of an example of the current multiplication circuitry of FIGS. 1, 2, 3, and 4.

[0012] FIG. 6 is a flowchart representative of example operations that may be at least one of executed, instantiated, or performed to implement the current multiplication circuitry of FIGS. 1, 2, 3, 4, and 5 to regulate a slew-rate of a voltage across a transistor.

[0013] FIG. 7 is a flowchart representative of example operations that may be at least one of executed, instantiated, or performed to implement the gate driver circuitry of FIGS. 1, 2, 3, and 4 to dynamically adjust a slew-rate of a voltage across a transistor.

[0014] FIG. 8 is a timing diagram of example operations of the gate driver circuitry of FIGS. 1, 2, 3, and 4 to regulate the slew-rate of a voltage across a transistor.

[0015] The drawings are not necessarily to scale. Generally, the same reference numbers in the drawing(s) and this description refer to the same or similar (functionally and/or structurally) features and/or parts. Although the drawings show regions with clean lines and boundaries, some or all of these lines and boundaries may be idealized. In reality, the boundaries or lines may be unobservable, blended or irregular.

DETAILED DESCRIPTION

[0016] As electronics continue to advance, systems have become capable of safely operating at increasingly complex operating conditions, such as higher powers and higher speeds. In driver circuitry, increasingly complex circuitry implements advanced techniques for regulating a supply of power to a load. Such circuitry allows the driver circuitry to precisely control and regulate transistor switching despite complex operating conditions.

[0017] When driving direct current (DC) motors, driver circuitry connected to one terminal of the DC motor regulates a supply of power to the motor by controlling transistors, while the remaining terminal of the DC motor is connected to ground. A first transistor (referred to as a high-side transistor) controls a supply of power from a supply terminal to the motor and a second transistor (referred to as a low-side transistor) controls a supply of power from the motor to a common potential. During operations to drive (e.g., supply power to, cause motion of) the motor, the driver circuitry switches the transistors on and off to regulate a supply of power to the motor. In a first operation, the driver circuitry turns on the first transistor (e.g., enables, causes to conduct current) and turns off the second transistor (e.g., disables, causes to not conduct current). In a second operation, the driver circuitry turns off the first transistor and turns on the second transistor.

[0018] Switching between the first and second operations of the transistors allows the driver circuitry to regulate the supply of power to the motor. In some systems, modifying the ratio of time in the first operation opposed to time in the second operation allows the driver circuitry to control operations of the motor, such as speed, torque, etc. In such systems, the driver circuitry receives one or more control signals that use pulse width modulation (PWM) to control the timing of switching between configurations. The one or more control signals control the timing of switching using duty cycles, which represents a ratio of time that the driver circuitry is turned on (e.g., supplying power) over the time that the driver circuitry is turned off (e.g., not supplying power). Increasing the duty cycle of a control signal increases the power delivered to the motor, which

increases the speed or torque of the motor. Decreasing the duty cycle of the control signal decreases the power delivered to the motor, which decreases the speed or torque of the motor.

[0019] As transistor technologies continue to advance, the speeds at which transistors can switch continue to increase. As the switching speeds of transistors increase, the amount of time that a transistor takes to transition between being on and off decreases. Such transitions are characterized by a slew rate, which is a rate (e.g., volts per second) characterizing the change in voltage across the transistor. For example, when switching between a common potential (e.g., 0 volts, ground) and a four-hundred-volt supply, the slew rate characterizes the rate at which the drain-to-source voltage across a transistor either increases or decreases.

[0020] In some systems, driver circuitry sets the slew rate of the transistor by controlling a current at a gate terminal of the transistor (referred to as a gate drive current). The slew rate of the transistor increases as the gate drive current increases. However, in some systems, such as motor control, increasing the slew rate of the transistors increases the stress applied to components of the motor during switching. To regulate the gate drive current, which regulates the slew rate, some driver circuitry includes a capacitor (referred to as a high-voltage capacitor) having a relatively large capacitance between the gate and drain terminals of the transistor. To support the increasingly high voltage switching the capacitance of the capacitor continues to increase. The capacitor uses portions of the gate drive current to compensate for changes in the voltage of the transistor, which reduces the gate drive current. However, as switching voltages and currents continue to increase the capacitance of the capacitor continues to increase, which increases the system on chip (SoC) size of the driver circuitry.

[0021] Examples described herein include methods and apparatus to regulate transistor switching. In some described examples, gate driver circuitry generates and controls a gate drive current to regulate a slew rate of a voltage change across a transistor during a switching operation. The gate driver circuitry includes current multiplication circuitry that is coupled to the transistor by a relatively small capacitor. When switching begins, the gate driver circuitry supplies a first gate drive current to the gate of the transistor using first and second current source circuitry. Once the gate drive current enables the transistor (e.g., turns on, causes conduction), the capacitor begins to source current from the current multiplication circuitry responsive to a change in the voltage across the transistor. The current multiplication circuitry mirrors and scales the current being supplied to the capacitor. The current multiplication circuitry sinks the scaled current from the gate drive current, which mimics having a relatively large capacitance coupled between the gate and drain of the transistor. Advantageously, the current multiplication circuitry uses a capacitor having a relatively small capacitance to mimic having a relatively large capacitance coupled between the gate and drain of the transistor. Advantageously, the current multiplication circuitry prevents excessively high slew rates of the transistor from stressing a load by limiting the slew rate.

[0022] Also described herein, are examples where the gate driver circuitry includes first and second current source circuitry, charging circuitry, capacitor circuitry, and comparator circuitry. The first and second current source circuitry generate the gate drive current responsive to a control signal from external circuitry. The charging circuitry charges capacitors of the capacitor circuitry. In some examples, the capacitor circuitry includes a set of first capacitors and a set of second capacitors, which the charging circuitry is structured to charge. When charging, the set of capacitors generate a switching voltage. When not being charged, the set of capacitors supply a reference voltage. In example operations, the charging circuitry is structured to charge one of the set of capacitors and use the other one of the set of capacitors to supply the reference voltage. For example, the charging circuitry may charge the set of first capacitors to generate the switching voltage and use the set of second capacitors to supply the reference voltage. The comparator circuitry compares the switching voltage and the reference voltage. The comparator circuitry controls the second current source circuitry responsive to the comparison of the switching voltage to the reference voltage. In some examples, once the switching voltage becomes greater than the reference voltage, the comparator

circuitry disables the second current source circuitry, which decreases the gate drive current and the slew rate of the transistor.

[0023] Advantageously, the gate driver circuitry described herein includes circuitry to adjust the slew rate of the transistor during switching operations. Advantageously, using a relatively high slew rate increases power efficiency and using a relatively low slew rate decreases stress on a load. Advantageously, the gate driver circuitry initially sets the transistor to have a relatively high slew rate before decreasing the gate drive current to reduce the slew rate. Advantageously, switching between different slew rates increases power efficiency without excessively stressing the load.

[0024] FIG. 1 is a block diagram of an example drive system **100**. In the example of FIG. 1, the drive system **100** includes example driver circuitry **105** and an example motor **110**. The driver circuitry **105** of FIG. 1 includes example power stage circuitry **115**, example high-side gate driver circuitry **120**, and example low-side gate driver circuitry **125**. The low-side gate driver circuitry **125** of FIG. 1 includes first example current source circuitry **130**, second example current source circuitry **135**, an example capacitor **140**, example current multiplication circuitry **145**, example charging circuitry **150**, example capacitor circuitry **155**, example multiplexer circuitry **160**, example controller circuitry **165**, and example comparator circuitry **170**.

[0025] The driver circuitry **105** is coupled to the motor **110**. In some examples, the driver circuitry **105** is coupled to an external signal source, such as programmable circuitry, which is structured to supply one or more control signals. Examples of the driver circuitry **105** are further illustrated and described in connection with FIGS. 2, 3, 4, and 5, below. Although in the example of FIG. 1, the driver circuitry **105** includes the gate driver circuitry **120**, **125** and the power stage circuitry **115**, in other examples, the power stage circuitry **115** may be external to the driver circuitry **105**.

[0026] The motor **110** has a first terminal and a second terminal. The first terminal of the motor **110** is coupled to the driver circuitry **105**. The second terminal of the motor **110** is coupled to a common terminal, which supplies a common potential (e.g., ground). In some examples, the motor **110** is physically coupled to an external system. For example, in automotive systems, the motor **110** is mechanically coupled to a wheel. In industrial systems, the motor **110** is mechanically coupled to one or more components to drive operations, such as manufacturing. In the example of FIG. 1, the motor **110** is a direct current (DC) motor. Alternatively, the drive system **100** of FIG. 1 may be modified to replace the motor **110** with a stepper motor or alternative type of motor. Although in the example of FIG. 1, the driver circuitry **105** is structured to supply power to the motor **110**, the driver circuitry **105** may be coupled to any type of load.

[0027] The power stage circuitry **115** has a first terminal, a second terminal, a third terminal, a fourth terminal, and a fifth terminal. The first and second terminals of the power stage circuitry **115** are coupled to the gate driver circuitry **120**. The third and fourth terminals of the power stage circuitry **115** are coupled to the gate driver circuitry **125**. The fifth terminal of the power stage circuitry **115** is coupled to the motor **110**. Examples of the power stage circuitry **115** are illustrated and described in connection with FIGS. 2, 3, 4, and 5, below.

[0028] The gate driver circuitry **120** has a first terminal, a second terminal, and a third terminal. The first and second terminals of the gate driver circuitry **120** are coupled to the power stage circuitry **115**. The third terminal of the gate driver circuitry **120** may be coupled to external circuitry structured to supply a high-side control signal (PWM.sub.HI). The low-side control signal is a PWM signal that controls the switching of the power stage circuitry **115**.

[0029] The gate driver circuitry **125** has a first terminal, a second terminal, and a third terminal. The first and second terminals of the gate driver circuitry **125** are coupled to the power stage circuitry **115**. The third terminal of the gate driver circuitry **125** may be coupled to external circuitry that is structured to supply a low-side control signal (PWM.sub.LOW). The low-side control signal is a PWM signal that controls the switching of the power stage circuitry **115**. Examples of the gate driver circuitry **125** are illustrated and described in connection with FIGS. 2, 3, 4, and 5, below.

[0030] The current source circuitry **130** has a first terminal and a second terminal. The first terminal of the current source circuitry **130** is coupled to the current source circuitry **135**, the controller circuitry **165**, and may be coupled to external circuitry that is structured to supply the low-side control signal. The second terminal of the current source circuitry **130** is coupled to the power stage circuitry **115**, the current source circuitry **135**, and the current multiplication circuitry **145**. Examples of the current source circuitry **130** are illustrated and described in connection with FIGS. 2, 3, 4, and 5, below.

[0031] The current source circuitry **135** has a first terminal, a second terminal, and a control terminal. The first terminal of the current source circuitry **135** is coupled to the current source circuitry **130**, the controller circuitry **165**, and may be coupled to external circuitry that is structured to supply the low-side control signal. The second terminal of the current source circuitry **135** is coupled to the power stage circuitry **115**, the current source circuitry **130**, and the current multiplication circuitry **145**. The control terminal of the current source circuitry **135** is coupled to the comparator circuitry **170**. An example of the current source circuitry **135** is illustrated and described in connection with FIGS. 2, 3, and 4, below.

[0032] The capacitor **140** has a first terminal and a second terminal. The first terminal of the capacitor **140** is coupled to the power stage circuitry **115**. The second terminal of the capacitor **140** is coupled to the current multiplication circuitry **145** and the charging circuitry **150**. Examples of the capacitor **140** are illustrated and described in connection with FIGS. 2, 3, 4, and 5, below.

[0033] The current multiplication circuitry **145** has a first terminal, a second terminal, and a third terminal. The first terminal of the current multiplication circuitry **145** is coupled to the power stage circuitry **115** and the current source circuitry **130**, **135**. The second terminal of the current multiplication circuitry **145** is coupled to the charging circuitry **150**. The third terminal of the current multiplication circuitry **145** is coupled to the capacitor **140** and the charging circuitry **150**. Examples of the current multiplication circuitry **145** are illustrated and described in connection with FIGS. 2, 3, 4, and 5, below.

[0034] The charging circuitry **150** has a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, and a sixth terminal. The first terminal of the charging circuitry is coupled to the capacitor **140** and the current multiplication circuitry **145**. The second terminal of the charging circuitry **150** is coupled to the current multiplication circuitry **145**. The third and fourth terminals of the charging circuitry **150** are coupled to the capacitor circuitry **155**. The fifth and sixth terminals of the charging circuitry **150** are coupled to the multiplexer circuitry **160** and the controller circuitry **165**. An example of the charging circuitry is illustrated and described in connection with FIGS. 2, 3, and 4, below.

[0035] The capacitor circuitry **155** has a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, and a sixth terminal. The first and second terminals of the capacitor circuitry **155** are coupled to the charging circuitry **150**. The third, fourth, fifth, and sixth terminals of the capacitor circuitry **155** are coupled to the multiplexer circuitry **160**. An example of the capacitor circuitry **155** is illustrated and described in connection with FIGS. 2, 3, and 4, below.

[0036] The multiplexer circuitry **160** has a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, a seventh terminal, and an eighth terminal. The first, second, third, and fourth terminals of the multiplexer circuitry **160** are coupled to the capacitor circuitry **155**. The fifth and sixth terminals of the multiplexer circuitry **160** are coupled to the charging circuitry **150** and the controller circuitry **165**. The seventh and eighth terminals of the multiplexer circuitry **160** are coupled to the comparator circuitry **170**. An example of the multiplexer circuitry **160** and configurations of the multiplexer circuitry **160** are illustrated and described in connection with FIGS. 2, 3, and 4, below.

[0037] The controller circuitry **165** has a first terminal, a second terminal, and a third terminal. The first terminal of the controller circuitry **165** is coupled to the current source circuitry **130**, **135** and may be coupled to external circuitry that is structured to supply the low-side control signal. The

second and third terminals of the controller circuitry **165** are coupled to the charging circuitry **150** and the multiplexer circuitry **160**. An example of the controller circuitry **165** is illustrated and described in connection with FIG. 2, below.

[0038] The comparator circuitry **170** has a first terminal, a second terminal, and an output terminal. The first and second terminals of the comparator circuitry **170** are coupled to the multiplexer circuitry **160**. The output terminal of the comparator circuitry **170** is coupled to the current source circuitry **135**. An example of the comparator circuitry **170** is illustrated and described in connection with FIGS. 2, 3, and 4, below.

[0039] In example operations of the drive system **100**, the driver circuitry **105** receives the high-side and low-side control signals from external circuitry structured to control the driver circuitry **105**. In some examples, the external circuitry controls the operations of the motor **110** responsive to adjusting the control signals. For example, increasing the duty cycle of the high-side control signal and decreasing the duty cycle of the low-side control signals increases the amount of power supplied to the motor **110**. In such an example, the motor **110** supplies additional mechanical energy responsive to the increase in power from the driver circuitry **105**.

[0040] In example operations of the driver circuitry **105**, the gate driver circuitry **120**, **125** independently controls switching of the power stage circuitry **115**. In some examples, the power stage circuitry **115** supplies power to the motor **110** responsive to current from the high-side gate driver circuitry **120**. In such examples, the power stage circuitry **115** discharges energy from the motor **110** responsive to the low-side gate driver circuitry **125** sinking current from the motor **110**. In response to the gate driver circuitry **125** transitioning from supplying current to sinking current from the motor **110**, currents through the motor **110** rapidly change directions, which stresses electrical components of the motor **110**. In the example of FIG. 1, the gate driver circuitry **125** decreases the amount of stress put on the motor **110** during switching events using the current multiplication circuitry **145** and the current source circuitry **135** to control the gate drive current, which controls switching. Example operations of the gate driver circuitry **125** to reduce stress on the motor **110** and increase energy efficiency are described in connection with FIGS. 2, 3, 4, 5, 6, and 7, below.

[0041] FIG. 2 is a schematic diagram of an example drive system **200**, which is an example of the drive system **100** of FIG. 1. In the example of FIG. 2, the drive system **200** includes example driver circuitry **202** and an example load **204**. The example driver circuitry **202** of FIG. 2 includes example power stage circuitry **206**, example high-side gate driver circuitry **208**, and example low-side gate driver circuitry **210**. The example power stage circuitry **206** of FIG. 2 includes a first example transistor **214** and a second example transistor **216**. The example low-side gate driver circuitry **210** of FIG. 2 includes first example current source circuitry **218**, second example current source circuitry **220**, a first example capacitor **222**, example current multiplication circuitry **224**, example charging circuitry **226**, example capacitor circuitry **228**, example multiplexer circuitry **230**, example controller circuitry **232**, and example comparator circuitry **233**.

[0042] The example current multiplication circuitry **224** of FIG. 2 includes a third example transistor **234**, a fourth example transistor **236**, a fifth example transistor **238**, and a sixth example transistor **240**. The example charging circuitry **226** of FIG. 2 includes a seventh example transistor **242**, a first example switch **244**, an eighth example transistor **246**, and a second example switch **248**. The example capacitor circuitry **228** of FIG. 2 includes a second example capacitor **250**, a third example capacitor **252**, a fourth example capacitor **254**, and a fifth example capacitor **256**. The example multiplexer circuitry **230** of FIG. 2 includes a third example switch **258**, a fourth example switch **260**, a fifth example switch **262**, and a sixth example switch **264**. The example controller circuitry **232** of FIG. 2 includes a first example inverter **266**, an example buffer **268**, an example flip-flop **270**, and a second example inverter **272**.

[0043] The driver circuitry **202** is coupled to the load **204**. The driver circuitry **202** may be coupled to external circuitry structured to supply the high-side and low-side control signals. The driver

circuitry **202** is an example of the driver circuitry **105** of FIG. **1**. The load **204** is coupled to the driver circuitry **202**. In some examples, the load **204** may be mechanically or electrically coupled to one or more components. In some examples, the load **204** represents the motor **110** of FIG. **1**. [0044] The power stage circuitry **206** is coupled to the load **204** and the gate driver circuitry **208**, **210**. The power stage circuitry **206** is an example of the power stage circuitry **115** of FIG. **1**. The gate driver circuitry **208** is coupled to the power stage circuitry **206** and may be coupled to external circuitry that is structured to supply the high-side supply signal. The gate driver circuitry **208** is an example of the gate driver circuitry **120** of FIG. **1**. The gate driver circuitry **210** is coupled to the power stage circuitry **206** and may be coupled to external circuitry that is structured to supply the low-side supply signal. The gate driver circuitry **210** is an example of the gate driver circuitry **125** of FIG. **1**.

[0045] The transistor **214** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **214** is coupled to a first supply terminal, which supplies a first supply voltage ($V_{sub.SUP_1}$). The second and third terminals of the transistor **214** are coupled to the load **204**, the gate driver circuitry **210**, and the transistor **216**. In some examples, the third terminal of the transistor **214** is referred to as a bulk terminal. The control terminal of the transistor **214** is coupled to the gate driver circuitry **208**. In some examples, the transistor **214** is referred to as a high-side transistor, which supplies the first supply voltage to the load **204** based on a high-side drive current from the gate driver circuitry **208**.

[0046] The transistor **216** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **216** is coupled to the load **204**, the gate driver circuitry **210**, and the transistor **214**. The second and third terminals of the transistor **216** are coupled to a common terminal, which supplies a common potential (e.g., ground, AVSS, etc.). In some examples, the third terminal of the transistor **214** is referred to as the bulk terminal. The control terminal of the transistor **216** is coupled to the gate driver circuitry **210**. In some examples, the transistor **216** is referred to as a low-side transistor, which provides a current path to the common potential based on a low-side drive current from the gate driver circuitry **210**.

[0047] The current source circuitry **218** is coupled to the power stage circuitry **206**, the current source circuitry **220**, the current multiplication circuitry **224**, the controller circuitry **232**, and may be coupled to the external circuitry that is structured to supply the low-side control signal. The current source circuitry **218** is an example of the current source circuitry **130** of FIG. **1**. The current source circuitry **220** is coupled to the power stage circuitry **206**, the current source circuitry **218**, the current multiplication circuitry **224**, the controller circuitry **232**, the comparator circuitry **233**, and may be coupled to the external circuitry that is structured to supply the low-side control signal. The current source circuitry **220** is an example of the current source circuitry **135** of FIG. **1**.

[0048] The capacitor **222** has a first terminal and a second terminal. The first terminal of the capacitor **222** is coupled to the load **204** and the transistors **214**, **216**. The second terminal of the capacitor **222** is coupled to the current multiplication circuitry **224** and the charging circuitry **226**. The capacitor **222** is an example of the capacitor **140** of FIG. **1**. The current multiplication circuitry **224** is coupled to the power stage circuitry **206**, the current source circuitry **218**, **220**, the capacitor **222**, and the charging circuitry **226**. The current multiplication circuitry **224** is an example of the current multiplication circuitry **145** of FIG. **1**. Another example of the current multiplication circuitry **145**, **224** is illustrated and described in connection with FIG. **5**, below.

[0049] In some examples, the capacitor **222** has a capacitance that is proportional to a gate-to-drain capacitance (C_{gd}) of the transistor **216**. In such examples, the current multiplication circuitry **224** uses the capacitor **222** to compensate for the transistor **216** having a relatively low gate-to-drain capacitance, such as when the transistor **216** is a gallium nitride substrate (GaN) transistor. Advantageously, the current multiplication circuitry **224** compensates for a relatively low gate-to-drain capacitance of the transistor **216** using the capacitor **222**.

[0050] The charging circuitry **226** is coupled to the capacitor **222**, the current multiplication

circuitry **224**, the capacitor circuitry **228**, the multiplexer circuitry **230**, and the controller circuitry **232**. The charging circuitry **226** is an example of the charging circuitry **150** of FIG. **1**. The capacitor circuitry **228** is coupled to the charging circuitry **226** and the multiplexer circuitry **230**. The capacitor circuitry **228** is an example of the capacitor circuitry **155** of FIG. **1**. The multiplexer circuitry **230** is coupled to the charging circuitry **226**, the capacitor circuitry **228**, the controller circuitry **232**, and the comparator circuitry **233**. The multiplexer circuitry **230** is an example of the multiplexer circuitry **160** of FIG. **1**. The controller circuitry **232** is coupled to the current source circuitry **130**, **135**, the charging circuitry **150**, the multiplexer circuitry **160**, and may be coupled to the external circuitry that is structured to supply the low-side control signal. The controller circuitry **232** is an example of the controller circuitry **165** of FIG. **1**. The comparator circuitry **233** is coupled to the current source circuitry **220** and the multiplexer circuitry **230**. The comparator circuitry **233** is an example of the comparator circuitry **170** of FIG. **1**.

[0051] The transistor **234** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor **234** are coupled to a second supply terminal, which supplies a second supply voltage ($V_{\text{sub.SUP_2}}$). In some examples, the second supply voltage is less than the first supply voltage. In such examples, the second supply voltage allows the gate driver circuitry **210** to use relatively lower voltage transistors, which are capable of switching at higher speeds. In some examples, the second terminal of the transistor **234** is referred to as a bulk terminal. The third and control terminals of the transistor **234** are coupled to the capacitor **222**, the charging circuitry **226**, and the transistor **236**. In the example of FIG. **2**, the transistor **234** is structured as diode circuitry, which regulates the direction of current supplied to the capacitor **222**. In some examples, the transistor **234** is illustrated and described as diode circuitry. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate the transistor **234** using diode circuitry.

[0052] The transistor **236** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor **236** are coupled to the second supply terminal, which supplies the second supply voltage. In some examples, the second terminal of the transistor **236** is referred to as a bulk terminal. The third terminal of the transistor **236** is coupled to the transistors **238**, **240**. The control terminal of the transistor **236** is coupled to the capacitor **222**, the charging circuitry **226**, and the transistor **234**. In the example of FIG. **2**, the transistor **236** is structured as current mirror circuitry. In some examples, the transistor **236** mirrors the current flowing through the transistor **234**. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate one or both of the transistors **234**, **236** as current mirror circuitry.

[0053] The transistor **238** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and control terminals of the transistor **238** are coupled to the transistors **236**, **240**. The second and third terminals of the transistor **238** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **238** is referred to as a bulk terminal. In the example of FIG. **2**, the transistor **238** is structured as current mirror circuitry. In some examples, the transistor **238** mirrors the current flowing through the transistor **236**. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate the transistor **238** as current mirror circuitry.

[0054] The transistor **240** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **240** is coupled to the transistor **216** and the current source circuitry **218**, **220**. The second and third terminals of the transistor **240** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **240** is referred to as a bulk terminal. The control terminal of the transistor **240** is coupled to the transistors **236**, **238**. In the example of FIG. **2**, the transistor **240** is structured as current scaling circuitry. In some examples, the transistor **240** scales the current flowing through the transistor **238**. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate one or both of the transistors **238**, **240** as current scaling circuitry.

[0055] The transistor **242** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor **242** are coupled to the second supply terminal, which supplies the second supply voltage. In some examples, the second terminal of the transistor **242** is referred to as a bulk terminal. The third terminal of the transistor **242** is coupled to the switch **244**. The control terminal of the transistor **242** is coupled to the capacitor **222**, the current multiplication circuitry **224**, and the transistor **246**. In the example of FIG. 2, the transistor **242** is structured as control circuitry that, when enabled, supplies current to the switch **244**. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate the transistor **242** as control circuitry.

[0056] The switch **244** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **244** is coupled to the transistor **242**. The second terminal of the switch **244** is coupled to the capacitor circuitry **228**. The control terminal of the switch **244** is coupled to the multiplexer circuitry **230** and the controller circuitry **232**.

[0057] The transistor **246** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor **246** are coupled to the second supply terminal, which supplies the second supply voltage. In some examples, the second terminal of the transistor **246** is referred to as a bulk terminal. The third terminal of the transistor **246** is coupled to the capacitor circuitry **228**. The control terminal of the transistor **246** is coupled to the capacitor **222**, the current multiplication circuitry **224**, and the transistor **242**. In the example of FIG. 2, the transistor **246** is structured as control circuitry that, when enabled, supplies current to the switch **248**. Alternatively, the current multiplication circuitry **224** may be modified to replace or illustrate the transistor **246** as control circuitry.

[0058] The switch **248** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **248** is coupled to the transistor **246**. The second terminal of the switch **248** is coupled to the capacitor circuitry **228**. The control terminal of the switch **248** is coupled to the multiplexer circuitry **230** and the controller circuitry **232**. In some examples, the switches **244**, **248** are implemented using transistors. In other examples, the charging circuitry **226** may be modified to implement the switches **244**, **248** as switching circuitry or in combination with the transistors **242**, **246**.

[0059] The capacitor **250** has a first terminal and a second terminal. The first terminal of the capacitor **250** is coupled to the charging circuitry **226** and the multiplexer circuitry **230**. The second terminal of the capacitor **250** is coupled to the charging circuitry **226**, the multiplexer circuitry **230**, and the capacitor **254**. The capacitor **252** has a first terminal and a second terminal. The first terminal of the capacitor **252** is coupled to the charging circuitry **226** and the multiplexer circuitry **230**. The second terminal of the capacitor **252** is coupled to the charging circuitry **226**, the multiplexer circuitry **230**, and the capacitor **256**.

[0060] The capacitor **254** has a first terminal and a second terminal. The first terminal of the capacitor **254** is coupled to the multiplexer circuitry **230** and the capacitor **250**. The second terminal of the capacitor **254** is coupled to the common terminal, which supplies the common potential. The capacitor **256** has a first terminal and a second terminal. The first terminal of the capacitor **256** is coupled to the multiplexer circuitry **230** and the capacitor **252**. The second terminal of the capacitor **256** is coupled to the common terminal, which supplies the common potential.

[0061] In the example of FIG. 2, the capacitors **250**, **254** are referred to as a set of first capacitors **250**, **254** and the capacitors **252**, **256** are referred to as a set of second capacitors **252**, **256**. When the charging circuitry **226** is structured to charge (as further described below) the set of first capacitors **250**, **254**, the set of first capacitors **250**, **254** generate a switching voltage at the first terminal of the capacitor **250**. When the charging circuitry **226** is not structured to charge (as further described below) the set of first capacitors **250**, **254**, the set of first capacitors **250**, **254** supply a reference voltage at the second terminal of the capacitor **250** and the first terminal of the

capacitor **254**.

[0062] Similarly, when the charging circuitry **226** is structured to charge the set of second capacitors **252, 256**, the set of second capacitors **252, 256** generate the switching voltage at the first terminal of the capacitor **252**. Also, when the charging circuitry **226** is structured to discharge the set of second capacitors **252, 256**, the set of second capacitors **252, 256** generate the reference voltage at the second terminal of the capacitor **252** and the first terminal of the capacitor **256**.

Advantageously, the charging circuitry **226** may cause either the set of first capacitors **250, 254** or the set of second capacitors **252, 256** to generate either one of the reference voltage or the switching voltage. Advantageously, the reference voltage represents a discharge of the capacitor circuitry **228** and the switching voltage represents a charging of the capacitor circuitry **228**.

[0063] The switch **258** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **258** is coupled to the charging circuitry **226** and the capacitor circuitry **228**. The second terminal of the switch **258** is coupled to the switch **260** and the comparator circuitry **233**. The control terminal of the switch **258** is coupled to the charging circuitry **226**, the controller circuitry **232**, and the switch **262**. The switch **258** is structured to receive the switching voltage from the set of second capacitors **252, 256**.

[0064] The switch **260** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **260** is coupled to the charging circuitry **226** and the capacitor circuitry **228**. The second terminal of the switch **260** is coupled to the switch **258** and the comparator circuitry **233**. The control terminal of the switch **260** is coupled to the charging circuitry **226**, the controller circuitry **232**, and the switch **264**. The switch **260** is structured to receive the switching voltage from the set of first capacitors **250, 254**.

[0065] The switch **262** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **262** is coupled to the capacitor circuitry **228**. The second terminal of the switch **262** is coupled to the switch **264** and the comparator circuitry **233**. The control terminal of the switch **262** is coupled to the charging circuitry **226**, the controller circuitry **232**, and the switch **258**. The switch **262** is structured to receive the reference voltage from the set of first capacitors **250, 254**.

[0066] The switch **264** has a first terminal, a second terminal, and a control terminal. The first terminal of the switch **264** is coupled to the capacitor circuitry **228**. The second terminal of the switch **264** is coupled to the switch **262** and the comparator circuitry **233**. The control terminal of the switch **264** is coupled to the charging circuitry **226**, the controller circuitry **232**, and the switch **260**. The switch **264** is structured to receive the reference voltage from the set of second capacitors **252, 256**.

[0067] The inverter **266** has a first terminal and a second terminal. The first terminal of the inverter **266** is coupled to the current source circuitry **218, 220** and may be coupled to the external circuitry that is structured to supply the low-side control signal. The second terminal of the inverter **266** is coupled to the flip-flop **270**. In the example of FIG. 2, the inverter **266** is structured to invert the low-side control signal. Alternatively, the controller circuitry **232** may be modified to receive an inverted replica of the low-side control signal. In such an example, the controller circuitry **232** may be modified to remove the inverter **266**. The buffer **268** has a first terminal and a second terminal. The first terminal of the buffer **268** is coupled to a reset terminal, which supplies a reset signal. The second terminal of the buffer **268** is coupled to the flip-flop **270**. In the example of FIG. 2, the buffer **268** is structured to have propagation delay approximately equal to a propagation delay of the inverter **266**.

[0068] The flip-flop **270** has a clock terminal (CLK), a clear terminal (CLR), a data terminal (D), an output terminal (Q), and an inverted output terminal (QZ). The clock terminal of the flip-flop **270** is coupled to the inverter **266**. The clear terminal of the flip-flop **270** is coupled to the buffer **268**. The data terminal of the flip-flop **270** is coupled to the inverted output terminal of the flip-flop **270**. The output terminal of the flip-flop **270** is coupled to the charging circuitry **226**, the

multiplexer circuitry **230**, and the inverter **272**. In the example of FIG. 2, the flip-flop **270** is a data (D) flip-flop. Alternatively, the controller circuitry **232** may be modified to use an alternative type of flip-flop. The flip-flop **270** is structured as divider circuitry, which generates a first switching signal by dividing the frequency of the low-side control signal. The flip-flop **270** uses the first switching signal to control the switches **248**, **258**, **262**.

[0069] The inverter **272** has a first terminal and a second terminal. The first terminal of the inverter **272** is coupled to the charging circuitry **226**, the multiplexer circuitry **230**, and the flip-flop **270**. The second terminal of the inverter **272** is coupled to the charging circuitry **226** and the multiplexer circuitry **230**. The inverter **272** is structured to generate a second switching signal by inverting the first switching signal from the flip-flop **270**. The inverter **272** uses the second switching signal to control the switches **244**, **260**, **264**.

[0070] During example operations of the drive system **200**, the gate driver circuitry **210** controls the transistor **216** responsive to receiving the low-side control signal. During example operations to turn on the transistor **216** (e.g., enable, conducting, etc.), the controller circuitry **232** uses the first and second switching signals to structure the charging circuitry **226** and the multiplexer circuitry **230** to support a multi-slew turn on of the transistor **216**. In some such example operations, the gate driver circuitry **210** uses a first current to drive the transistor **216** for a first duration of time. However, in response to the comparator circuitry **233** determining that the switching voltage from the capacitor circuitry **228** is greater than or equal to the reference voltage from the capacitor circuitry **228**, the comparator circuitry **233** disables the current source circuitry **220**. After such a determination, the gate driver circuitry **210** uses a second current, which is less than the first current, to drive the transistor **216**. Advantageously, during the first duration of time, the current through the transistor **216** has a relatively high slew rate, which increases power efficiency, and after the first duration the current through the transistor **216** has a relatively low slew rate, which decreases stress on components of the load **204**. Advantageously, initially using a relatively high slew rate improves power efficiency during switching operations.

[0071] FIG. 3 is a schematic diagram of a first configuration of the example drive system **200** of FIG. 2, which is an example of the drive system **100** of FIG. 1. In the example of FIG. 3, the drive system **200** is structured to supply a reference voltage using the set of first capacitors **250**, **254** of FIG. 2 and generate a switching voltage by charging the set of second capacitors **252**, **256** of FIG. 2. In such example operations, the controller circuitry **232** of FIG. 2 opens the switches **244**, **260**, **264** of FIG. 2 and closes the switches **248**, **258**, **262** of FIG. 2. When in the first configuration of FIG. 3, the comparator circuitry **233** of FIG. 2 controls the current source circuitry **220** of FIG. 2 responsive to a comparison of the reference voltage from the set of first capacitors **250**, **254** to the switching voltage from the set of second capacitors **252**, **256**.

[0072] FIG. 4 is a schematic diagram of a second configuration of the example drive system **200** of FIG. 2, which is an example of the drive system **100** of FIG. 1. In the example of FIG. 4, the drive system **200** is structured to generate a switching voltage by charging the set of first capacitors **250**, **254** of FIG. 2 and supply a reference voltage using the set of second capacitors **252**, **256** of FIG. 2. In such example operations, the controller circuitry **232** of FIG. 2 closes the switches **244**, **260**, **264** of FIG. 2 and opens the switches **248**, **258**, **262** of FIG. 2. When in the second configuration of FIG. 4, the comparator circuitry **233** of FIG. 2 controls the current source circuitry **220** of FIG. 2 responsive to a comparison of the switching voltage from the set of first capacitors **250**, **254** to the reference voltage from the set of second capacitors **252**, **256**.

[0073] In the example of FIGS. 2, 3, and 4, the transistors **214**, **216**, **238**, **240** are n-channel metal-oxide semiconductor field-effect transistors (MOSFETs). Alternatively, the transistors **214**, **216**, **238**, **240** may be n-channel field-effect transistors (FETs), n-channel insulated-gate bipolar transistors (IGBTs), n-channel junction field effect transistors (JFETs), NPN bipolar junction transistors (BJTs) or, with slight modifications, p-type equivalent devices. In the example of FIGS. 2, 3, and 4, the transistors **234**, **236**, **242**, **246** are p-channel MOSFETs. Alternatively, the

transistors **234, 236, 242, 246** may be p-channel FETs, p-channel IGBTs, p-channel JFETs, NPN BJTs, or, with slight modifications, N-type equivalent devices. The transistors **214, 216, 234, 236, 238, 240, 242, 246** may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other type of device structure transistors. Furthermore, the transistors **214, 216, 234, 236, 238, 240, 242, 246** may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0074] FIG. 5 is a schematic diagram of an example drive system **500**, which is another example of the drive systems **100, 200** of FIGS. 1, 2, 3, and 4, that includes another example of current multiplication circuitry **145, 224** of FIGS. 1, 2, 3, and 4. In the example of FIG. 5, the drive system **500** includes example driver circuitry **502** and an example load **504**. The driver circuitry **502** of FIG. 5 includes example power stage circuitry **506**, example high-side gate driver circuitry **508**, and example low-side gate driver circuitry **510**. The power stage circuitry **506** of FIG. 5 includes a first example transistor **514** and a second example transistor **516**.

[0075] The example low-side gate driver circuitry **510** of FIG. 5 includes example current source circuitry **518**, an example capacitor **522**, and example current multiplication circuitry **524**. The current multiplication circuitry **524** of FIG. 5 includes a first example resistor **532**, a third example transistor **534**, a fourth example transistor **536**, a fifth example transistor **538**, a sixth example transistor **540**, a seventh example transistor **542**, a second example resistor **543**, an eighth example transistor **544**, a first example diode **546**, a third example resistor **548**, a ninth example transistor **550**, a first example inverter **552**, a second example inverter **554**, a tenth example transistor **556**, a fourth example resistor **557**, an eleventh example transistor **558**, a fifth example resistor **560**, a twelfth example transistor **562**, a second example diode **564**, a thirteenth example transistor **566**, and a fourteenth example transistor **568**. In the example of FIG. 5, the gate driver circuitry **510** implements single slew rate switching using the current source circuitry **518**. Alternatively, the gate driver circuitry **510** may be modified to implement multi slew rate switching, as shown in FIGS. 2, 3, and 4.

[0076] The driver circuitry **502** is coupled to the load **504**. The driver circuitry **502** may be coupled to external circuitry that is structured to supply the high-side and low-side control signals. The driver circuitry **502** is an alternative example of the driver circuitry **105, 202** of FIGS. 1, 2, 3, and 4. In the example of FIG. 5, the driver circuitry **502** includes the current multiplication circuitry **524**, which further includes circuitry to reduce a quiescent current during times that the transistor **516** is off. The load **504** is coupled to the driver circuitry **502**. In some examples, the load **504** may be mechanically or electrically coupled to one or more components. The load **504** is another example of the motor **110** of FIG. 1 and the load **204** of FIGS. 2, 3, and 4.

[0077] The power stage circuitry **506** is coupled to the load **504** and the gate driver circuitry **508, 510**. The power stage circuitry **506** is another example of the power stage circuitry **115, 206** of FIGS. 1, 2, 3, and 4. The gate driver circuitry **508** is coupled to the power stage circuitry **506** and may be coupled to external circuitry that is structured to supply the high-side supply signal. The gate driver circuitry **508** is another example of the gate driver circuitry **120, 208** of FIGS. 1, 2, 3, and 4. The gate driver circuitry **510** is coupled to the power stage circuitry **506** and may be coupled to external circuitry that is structured to supply the low-side supply signal. The gate driver circuitry **510** is an example of the gate driver circuitry **125, 210** of FIGS. 1, 2, 3, and 4.

[0078] The transistor **514** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **514** is coupled to the first supply terminal, which supplies the first supply voltage ($V_{sub.SUP_1}$). The second and third terminals of the transistor **514** are coupled to the load **504**, the gate driver circuitry **510**, and the transistor **516**. In some examples, the third terminal of the transistor **514** is referred to as a bulk terminal. The control terminal of the transistor **514** is coupled to the gate driver circuitry **508**. In some examples, the transistor **514** is referred to as a high-side transistor, which supplies the supply voltage to the load

504 based on a high-side drive current from the gate driver circuitry **508**. The transistor **514** is an example of the transistor **214** of FIGS. 2, 3, and 4.

[0079] The transistor **516** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **516** is coupled to the load **504**, the gate driver circuitry **510**, and the transistor **514**. The second and third terminals of the transistor **516** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **514** is referred to as the bulk terminal. The control terminal of the transistor **516** is coupled to the gate driver circuitry **510**. In some examples, the transistor **516** is referred to as a low-side transistor, which provides a current path to the common potential based on a low-side drive current from the gate driver circuitry **510**. The transistor **516** is an example of the transistor **216** of FIGS. 2, 3, and 4.

[0080] The current source circuitry **518** is coupled to the power stage circuitry **506**, the current multiplication circuitry **524**, and may be coupled to external circuitry that is structured to supply the low-side control signal. The current source circuitry **518** is another example of the current source circuitry **130**, **218** of FIGS. 1, 2, 3, and 4. The capacitor **522** has a first terminal and a second terminal. The first terminal of the capacitor **522** is coupled to the load **504** and the transistors **514**, **516**. The second terminal of the capacitor **522** is coupled to the current multiplication circuitry **524**. The capacitor **522** is another example of the capacitors **140**, **222** of FIGS. 1, 2, 3, and 4.

[0081] The current multiplication circuitry **524** is coupled to the power stage circuitry **506**, the current source circuitry **518**, the capacitor **522**, and the external circuitry that is structured to supply the low-side control signal. The current multiplication circuitry **524** is another example of the current multiplication circuitry **145**, **224** of FIGS. 1, 2, 3, and 4. In the example of FIG. 5, the current multiplication circuitry **524** includes additional circuitry that reduces the quiescent current by disabling components when the transistor **516** is off.

[0082] In some examples, the capacitor **522** has a capacitance that is proportional to a non-ideal gate-to-drain capacitance of the transistor **516**. In such examples, the current multiplication circuitry **524** is structured to use the capacitor **522** to compensate for the transistor **516** having a relatively low gate-to-drain capacitance, such as when the transistor **516** is a gallium nitride substrate (GaN) transistor. Advantageously, the current multiplication circuitry **524** is structured to compensate for a relatively low gate-to-drain capacitance of the transistor **516** using a relatively small capacitance.

[0083] The resistor **532** has a first terminal and a second terminal. The first terminal of the resistor **532** is coupled to the second supply terminal, which supplies the second supply voltage ($V_{\text{sub.SUP}_2}$). The second terminal of the resistor **532** is coupled to the capacitor **522** and the transistors **534**, **536**. In some examples, the resistor **532** is structured as discharge circuitry, which prevents the control terminals of the transistors **534**, **536** from floating when no signal is present (also referred to as deadtime). Alternatively, the current multiplication circuitry **524** may be modified to exclude the resistor **532** or implement another method of discharging the control terminals of the transistors **534**, **536**.

[0084] The transistor **534** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor **534** are coupled to the second supply terminal, which supplies the second supply voltage. In some examples, the second terminal of the transistor **534** is referred to as a bulk terminal. The third and control terminals of the transistor **534** are coupled to the capacitor **522**, the resistor **532**, and the transistor **536**. In the example of FIG. 5, the transistor **534** is structured as diode circuitry, which regulates the direction of current supplied to the capacitor **522**. In some examples, the transistor **534** is illustrated and described as diode circuitry. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate the transistor **534** using diode circuitry.

[0085] The transistor **536** has a first terminal, a second terminal, a third terminal, and a control

terminal. The first and second terminals of the transistor **536** are coupled to the second supply terminal, which supplies the second supply voltage. In some examples, the second terminal of the transistor **536** is referred to as a bulk terminal. The third terminal of the transistor **536** is coupled to the transistors **538**, **540**, **542** and the resistor **543**. The control terminal of the transistor **536** is coupled to the capacitor **522**, the resistor **532**, and the transistor **534**. In the example of FIG. 5, the transistor **536** is structured as current mirror circuitry. In some examples, the transistor **536** mirrors the current flowing through the transistor **534**. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate one or both of the transistors **534**, **536** using current mirror circuitry.

[0086] The transistor **538** has a first terminal, a second terminal, a third terminal, and a control terminal. The first and control terminals of the transistor **538** are coupled to the transistors **536**, **540**, **542**, and the resistor **543**. The second and third terminals of the transistor **538** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **538** is referred to as a bulk terminal. In the example of FIG. 5, the transistor **538** is structured as current mirror circuitry. In some examples, the transistor **538** mirrors the current flowing through the transistor **536**. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate the transistor **538** using current mirror circuitry.

[0087] The transistor **540** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **540** is coupled to the transistor **516** and the current source circuitry **518**. The second and third terminals of the transistor **540** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **540** is referred to as a bulk terminal. The control terminal of the transistor **540** is coupled to the transistors **536**, **538**, **542** and the resistor **543**. In the example of FIG. 5, the transistor **540** is structured as current scaling circuitry. In some examples, the transistor **540** scales the current flowing through the transistor **538**. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate one or both of the transistors **538**, **540** using alternative current scaling circuitry.

[0088] The transistor **542** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **542** is coupled to the transistors **536**, **538**, **540** and the resistor **543**. The second and third terminal of the transistor **542** is coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **542** is referred to as a bulk terminal. The control terminal of the transistor **542** is coupled to the transistors **544**, **566** and the inverters **552**, **554**. In the example of FIG. 5, the transistor **542** is structured as control circuitry that, when enabled, prevents the transistors **538**, **540** from being turned on. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate the transistor **542** as control circuitry.

[0089] The resistor **543** has a first terminal and a second terminal. The first terminal of the resistor **543** is coupled to the transistors **536**, **538**, **540**, **542**. The second terminal of the resistor **543** is coupled to the common terminal, which supplies the common potential. In some examples, the resistor **543** is structured as discharge circuitry, which prevents the control terminals of the transistors **538**, **540** from floating when no signal is present (also referred to as deadtime). Alternatively, the current multiplication circuitry **524** may be modified to exclude the resistor **543** or implement another method of discharging the control terminals of the transistors **538**, **540**.

[0090] The transistor **544** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **544** is coupled to the second supply terminal, which supplies the second supply voltage. The second and third terminals of the transistor **544** are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor **544** is referred to as a bulk terminal. The control terminal of the transistor **544** is coupled to the transistors **542**, **566** and the inverters **552**, **554**. In the example of FIG. 5, the transistor **544** is structured as control circuitry that, when enabled, prevents the transistors **534**, **536**

from being turned on. Alternatively, the current multiplication circuitry 524 may be modified to replace or illustrate the transistor 544 with alternative control circuitry. In such examples, the transistor 544 may be illustrated as switching circuitry.

[0091] The diode 546 has a first terminal and a second terminal. The first terminal of the diode 546 is coupled to the secondary supply terminal, which supplies the second supply voltage. The second terminal of the diode 546 is coupled to the resistor 548 and the transistor 550. In the example of FIG. 5, the diode 546 is a Zener diode that is structured to breakdown (e.g., conduct current) responsive to a voltage difference greater than a threshold voltage. The resistor 548 has a first terminal and a second terminal. The first terminal of the resistor 548 is coupled to the diode 546 and the transistor 550. The second terminal of the resistor 548 is coupled to the common terminal, which supplies the common potential. The transistor 550 has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor 550 is coupled to the second supply terminal, which supplies the second supply voltage. The second and third terminals of the transistor 550 are coupled to the common terminal, which supplies the common potential. In some examples, the third terminal of the transistor 550 is referred to as a bulk terminal. The control terminal of the transistor 550 is coupled to the diode 546 and the resistor 548.

[0092] In the example of FIG. 5, the diode 546, the resistor 548, and the transistor 550 are structured as clamp circuitry. In some examples, the diode 546, the resistor 548, and the transistor 550 prevent the second supply voltage from increasing beyond a maximum voltage by clamping the second supply voltage. For example, when the second supply voltage exceeds a breakdown voltage of the diode 546, the resistor 548 uses current from the diode 546 to turn on the transistor 550, which clamps the second supply voltage. Alternatively, the current multiplication circuitry 524 may be modified to replace or illustrate one or more of the diode 546, the resistor 548, or the transistor 550 as clamp circuitry. In such examples, one or more of the diode 546, the resistor 548, or the transistor 550 may be illustrated as clamp circuitry.

[0093] The inverter 552 has a first terminal and a second terminal. The first terminal of the inverter 552 may be coupled to external circuitry that is structured to supply the low-side control signal. The second terminal of the inverter 552 is coupled to the transistors 542, 544, 566 and the inverter 554. The inverter 554 has a first terminal and a second terminal. The first terminal of the inverter 554 is coupled to the transistors 542, 544, 566 and the inverter 552. The second terminal of the inverter 554 is coupled to the transistor 556.

[0094] The transistor 556 has a first terminal, a second terminal, and a control terminal. The first terminal of the transistor 556 is coupled to the transistors 558, 562 and the resistor 560. The second terminal of the transistor 556 is coupled to the resistor 557. The control terminal of the transistor 556 is coupled to the inverter 554. The resistor 557 has a first terminal and a second terminal. The first terminal of the resistor 557 is coupled to the transistor 556. The second terminal of the resistor 557 is coupled to the common terminal, which supplies the common potential. In some examples, the resistor 557 is a current limiting resistor, which prevents excessively high currents from flowing through the transistors 556, 558. Alternatively, the current multiplication circuitry 524 may be modified to exclude the resistor 557 or implement another method of current regulation.

[0095] The transistor 558 has a first terminal, a second terminal, a third terminal, and a control terminal. The first and second terminals of the transistor 558 are coupled to the current source circuitry 518, the resistor 560, the transistors 562, 568, and external circuitry that is structured to supply the low-side control signal. In some examples, the second terminal of the transistor 558 is referred to as a bulk terminal. The third and control terminals of the transistor 558 are coupled to the transistors 556, 562 and the resistor 560. The resistor 560 has a first terminal and a second terminal. The first terminal of the resistor 560 is coupled to the current source circuitry 518, the transistors 558, 562, 568, and external circuitry that is structured to supply the low-side control signal. The second terminal of the resistor 560 is coupled to the transistors 556, 558, 562. The transistor 562 has a first terminal, a second terminal, a third terminal, and a control terminal. The

first terminal and second terminals of the transistor **562** are coupled to the current source circuitry **518**, the transistors **558**, **568**, the resistor **560**, and the external circuitry structured to supply the secondary supply voltage.

[0096] In the example of FIG. 5, the transistor **558**, **562** are structured as current mirror circuitry. In some examples, the transistor **562** mirrors the current flowing through the transistor **558**.

Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate one or both of the transistors **558**, **562** using current mirror circuitry. In such examples, the transistors **558**, **562** may be illustrated as current mirror circuitry.

[0097] The diode **564** has a first terminal and a second terminal. The first terminal of the diode **564** is coupled to the transistors **562**, **566**, **568**. The second terminal of the diode **564** is coupled to the common terminal, which supplies the common potential. In the example of FIG.

[0098] **5**, the diode **564** is a Zener diode that is structured to breakdown (e.g., conduct current) responsive to a voltage difference greater than a threshold voltage.

[0099] The transistor **566** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **566** is coupled to the transistors **562**, **568** and the diode **564**. The second and third terminals of the transistor **566** are coupled to the common terminal, which supplies the common potential. The control terminal of the transistor **566** is coupled to the transistors **542**, **544** and the inverters **552**, **554**. In the example of FIG. 5, the transistor **566** is structured as control circuitry that, when enabled, prevents the transistor **568** from being turned on. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate the transistor **566** as control circuitry. In such examples, the transistor **566** may be illustrated as switching circuitry.

[0100] The transistor **568** has a first terminal, a second terminal, a third terminal, and a control terminal. The first terminal of the transistor **568** is coupled to the auxiliary supply terminal, which supplies the auxiliary supply voltage. The second and third terminals of the transistor **568** are coupled to the second supply terminal, which supplies the second supply voltage. The control terminal of the transistor **568** is coupled to the transistors **562**, **566** and the diode **564**. In the example of FIG. 5, the transistor **568** is structured as regulator circuitry. In some examples, the transistor **568** regulates the auxiliary supply voltage to set the second supply voltage. Alternatively, the current multiplication circuitry **524** may be modified to replace or illustrate the transistor **568** as regulator circuitry. In such examples, the transistor **568** may be illustrated as supply circuitry.

[0101] In the example of FIG. 5, the transistors **514**, **516**, **538**, **540**, **542**, **544**, **550**, **556**, **566**, **568** are n-channel MOSFETs. Alternatively, the transistors **514**, **516**, **538**, **540**, **542**, **544**, **550**, **556**, **566**, **568** may be n-channel FETs, n-channel IGBTs, n-channel JFETs, NPN BJTs. In the example of FIG. 5, the transistors **534**, **536**, **558**, **562** are p-channel MOSFETs. Alternatively, the transistors **534**, **536**, **558**, **562** may be p-channel FETs, p-channel IGBTs, p-channel JFETs, NPN BJTs. One or more of the transistors **514**, **516**, **534**, **536**, **538**, **540**, **542**, **544**, **550**, **556**, **566**, **558**, **562** may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other type of device structure transistors. Furthermore, the transistors **514**, **516**, **534**, **536**, **538**, **540**, **542**, **544**, **550**, **556**, **566**, **558**, **562** may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0102] FIG. 6 is a flowchart representative of example operations **600** that may be at least one of executed, instantiated, or performed to implement the current multiplication circuitry **145**, **224**, **524** of FIGS. 1, 2, 3, 4, and 5 to regulate a slew-rate of a voltage across a transistor (e.g., the transistors **216**, **516** of FIGS. 2, 3, 4, and 5). The example operations **600** of FIG. 6 begin at Block **610** at which the gate driver circuitry **125**, **210**, **510** of FIGS. 1, 2, 3, 4, and 5 receives a switching signal to turn on a transistor. (Block **610**). In some examples, external circuitry supplies a low-side control signal to the gate driver circuitry **125**, **210**. In such examples, the low-side control signal is a pulse width modulation (PWM) signal having a duty cycle, which determines an on time and an off time of the transistors **216**, **516**. The on time corresponds to durations of time during which the gate

driver circuitry **125, 210, 510** supplies a drive current that causes the transistors **216, 516** to conduct current. The off time corresponds to durations of time where the gate driver circuitry **125, 210, 510** does not generate a drive current capable of causing the transistor **216, 516** to conduct current. In such example operations, adjusting the duty cycle of the low-side control signal allows the external circuitry to modify a supply of power to the motor **110** of FIG. 1.

[0103] The current source circuitry **130, 135, 218, 220, 518** of FIGS. 1, 2, 3, 4, and 5 generate a gate drive current using the switching signal. (Block **620**). In some examples, the current source circuitry **130, 135, 218, 220, 518** supply current responsive to receiving the low-side control signal corresponding to the on time. For example, the current source circuitry **130, 135, 218, 220, 518** supply current responsive to receiving a low-side control signal that is a logic high (e.g., logical one). In such examples, currents from the current source circuitry **130, 135, 218, 220, 518** combine to generate the gate drive current. However, as described in FIG. 7, below, disabling the current source circuitry **135, 220** sets the gate drive current equal to the current from the current source circuitry **130, 218**. Advantageously, the gate driver circuitry **125, 210** may modify the gate driver current using the current source circuitry **135, 220**.

[0104] The current source circuitry **130, 135, 218, 220, 518** set a slew rate of voltage changes across the transistor. (Block **630**). In some examples, the current source circuitry **130, 135, 218, 220, 518** set the slew rate of the transistors **216, 516** responsive to the magnitude of the gate driver current. For example, the transistors **216, 516** have a higher slew rate when the gate driver current is increased and has a lower slew rate when the gate driver current is decreased. In such examples, the current source circuitry **130, 135, 218, 220, 518** are structured to supply currents that set the slew rate of the transistors **216, 516** to a designed value. Advantageously, adjusting the gate drive current adjusts the slew rate of the transistors **216, 516**. Advantageously, the current source circuitry **130, 135, 218, 220** may be structured to set the slew rate of the transistors **216, 516**.

[0105] The current multiplication circuitry **145, 224, 524** supplies a reference current to a capacitor responsive to changes in the voltage of a drain of the transistor. (Block **640**). In some examples, the transistors **234, 534** of FIGS. 2, 3, 4, and 5 are structured as diode circuitry, which supply current to the capacitors **222, 522** of FIGS. 2, 3, 4, and 5. In example operation, once the current source circuitry **130, 135, 218, 220, 518** supplies a gate drive current to the transistors **216, 516**, the voltage across the capacitors **222, 522** begins to change. The capacitors **222, 522** source current from the transistors **234, 534** responsive to changes in the voltage across the transistors **216, 516**. Advantageously, the operations of the capacitor **222, 522** using the transistors **234, 534** are proportional to having a capacitor coupled between the drain and gate terminals of the transistors **216, 516**.

[0106] The current multiplication circuitry **145, 224, 524** mirrors the reference current. (Block **650**). In some examples, the transistors **236, 536** of FIGS. 2, 3, 4, and 5 are structured as current mirror circuitry, which mirror the current through the transistors **234, 534**. In such examples, the transistors **236, 536** supply the replica of the reference current to the transistors **238, 240, 538, 540** of FIGS. 2, 3, 4, and 5. Advantageously, the transistors **234, 534** mirror the current being supplied to the capacitors **222, 522**.

[0107] The current multiplication circuitry **145, 224, 524** scales the reference current. (Block **660**). In some examples, the transistors **238, 538** are structured as current mirror circuitry, which sinks a current approximately equal to the replica of the reference current from the transistors **236, 536**. Also, the transistors **238, 538** control the transistors **240, 540** responsive to sinking the replica of the reference current. In some examples, the transistors **240, 540** are structured as scaling circuitry, which scales the current flowing through the transistors **238, 538**. In such examples, the transistors **240, 540** are sized (e.g., having characteristics that scale) in relation to the transistors **238, 538**. In example operations, the transistors **240, 540** sink a current approximately equal to the ratio of the size of the transistors **240, 540** to the transistors **238, 538**. For example, the transistors **240, 540** may sink a current that is approximately sixteen times the current flowing through the transistors

238, 538.

[0108] The current multiplication circuitry **145, 224, 524** sinks the scaled reference current from the gate drive current to regulate the slew rate of voltage changes across the transistor. (Block **670**). In some examples, the transistors **240, 540** are structured as current sink circuitry, which sinks the scaled current from the control terminal of the transistor **216**. In such examples, the transistors **240, 540** decrease the gate drive current responsive to sinking the scaled current.

[0109] Advantageously, the current multiplication circuitry **145, 224, 524** uses the scaled current to mimic having a relatively large capacitance coupled between the gate and drain terminals of the transistors **216, 516**. Advantageously, the current multiplication circuitry **145, 224, 524** decreases the size of a capacitance needed to regulate the slew rate of the transistors **216, 516**. For example, when an eight picofarad capacitance is needed between the gate and drain terminals of the transistors **216, 516** and the current multiplication circuitry **145, 224, 524** scales the reference current by sixteen, the capacitors **140, 222, 522** may have a capacitance of half of a picofarad (pF). In such examples, the capacitors **140, 222, 522** and the current multiplication circuitry **145, 224, 524** mimic the eight picofarad capacitance using a capacitance that is one-sixteenth the size. Advantageously, decreasing the capacitance of the capacitors **140, 222, 522** decreases the system on chip (SoC) size of the gate driver circuitry **125, 210, 510**. Advantageously, in GaN designs, decreasing the capacitance of the capacitors **140, 222, 522** decreases integration complexity of the gate driver circuitry **210, 510**.

[0110] In some examples, the gate driver circuitry **125, 210, 510** dynamically adjusts the slew rate. (Blocks **710, 720, 730, 740, 750, 760, 770, 780, 790** of FIG. 7). Example operations of Blocks **710, 720, 730, 740, 750, 760, 770, 780, 790** are illustrated and described in connection with FIG. 7, below. Advantageously, dynamically adjusting the slew rate of the transistors **216, 516** allows the gate driver circuitry **125, 210, 510** to increase power efficiency by first using a relatively high slew rate, then using a relatively low slew rate. Advantageously, dynamically adjusting the slew rate of the transistors **216, 516** increases power efficiency and reduces stress on the motor **110**.

[0111] Although example methods are described with reference to the flowchart illustrated in FIG. 6, many other methods of implementing the current multiplication circuitry **145, 224, 524** of FIGS. 1, 2, 3, 4, and 5 may also be used in this description. For example, the order of execution of the blocks may be changed, or some of the blocks described may be changed, eliminated, or combined. Similarly, additional operations may be included in the manufacturing process before, in between, or after the blocks shown in the illustrated examples.

[0112] FIG. 7 is a flowchart representative of example operations **700** that may be at least one of executed, instantiated, or performed to implement the gate driver circuitry **125, 210** of FIGS. 1, 2, 3, and 4 to dynamically adjust a slew-rate of a voltage across a transistor (e.g., the transistor **216** of FIGS. 2, 3, and 4). The example operations **700** of FIG. 7 begin with Blocks **610, 620, 630, 640, 650, 660, 670** of FIG. 6 at which the gate driver circuitry **125, 210** sets a slew rate of a voltage change across a transistor using a driver current. (Blocks **610, 620, 630, 640, 650, 660, 670**). Example operations of Blocks **610, 620, 630, 640, 650, 660, 670** are illustrated and described in connection with FIG. 6, above.

[0113] The controller circuitry **165, 232** of FIGS. 1 and 2 determines if this is a first switching cycle. (Block **710**). In some examples, the first switching cycle is a first pulse of the low-side control signal after at least one of a power-up or a reset. In such examples, the set of first capacitors **250, 254** of FIGS. 2, 3, and 4 or the set of second capacitors **252, 256** of FIGS. 2, 3, and 4 of the capacitor circuitry **155, 228** of FIGS. 1, 2, 3, and 4 are charged. The controller circuitry **165, 232** may include additional circuitry to determine whether the switching cycle is a first switching cycle. For example, the controller circuitry **165, 232** may include an additional flip-flop that is set the first cycle after a power-up or a reset. During the first switching cycle, the comparator circuitry **170, 233** of FIGS. 1, 2, 3, and 4 compares a switching voltage from one of the set of first capacitors **250, 254** or the set of second capacitors **252, 256**. The one of the set of first capacitors **250, 254** or the

set of second capacitors **252, 256** generates the switching voltage responsive to being charged. However, prior to the first switching cycle, the capacitors **250, 252, 254, 256** have not been charged.

[0114] If the controller circuitry **165, 232** determines that this is a first switching cycle (e.g., Block **710** returns a result of YES), the charging circuitry **150, 226** of FIGS. **1, 2, 3,** and **4** charges first capacitors. (Block **720**). In some examples, during the first switching cycle, the controller circuitry **165, 232** of FIGS. **1** and **2** closes the switch **244** of FIGS. **2, 3,** and **4** to charge the set of first capacitors **250, 254**. Also, during the first switching cycle, the controller circuitry **165, 232** opens the switch **248** of FIGS. **2, 3,** and **4** to keep the set of second capacitors **252, 256** discharged. In such examples, the charging circuitry **226** may be referred to as being structured to charge the set of first capacitors **250, 254** and the charging circuitry **226** may be referred to as being structured not to charge the set of second capacitors **252, 256**. Control proceeds to return to Block **610**.

[0115] If the controller circuitry **165, 232** determines that this is not a first switching cycle (e.g., Block **710** returns a result of NO), the controller circuitry **165, 232** determines if the switching cycle is an even switching cycle. (Block **730**). In some examples, the controller circuitry **165, 232** generates a first and second switching signals to control the switches **244, 248, 258, 260, 262, 264** of FIGS. **2, 3,** and **4**. In such examples, the flip-flop **270** of FIG. **2** generates the first switching signal by dividing the frequency of the low-side control signal and the inverter **272** of FIG. **2** generates the second switching signal by inverting the first switching signal. The controller circuitry **165, 232** uses the first switching signal to control the switch **244**, which is structured to charge the set of first capacitors **250, 254** when closed. In example operation, the flip-flop **270** switches the states of the first and second switching signals after each pulse of the low-side control signal. For example, during even pulses (e.g., every other pulse), the first switching signal is in a first state and the second switching signal is in a second state. In such examples, during odd pulses (e.g., every other pulse), the first switching signal is in the second state and the second switching signal is in the first state.

[0116] If the controller circuitry **165, 232** determines that the switching cycle is an even switching cycle (e.g., Block **730** returns a result of YES), the charging circuitry **150, 220** charges second capacitors to generate a switching voltage. (Block **740**). In some examples, the controller circuitry **165, 232** causes the charging circuitry **150, 226** to charge the set of second capacitors **252, 256** responsive to the second switching signal closing the switch **248**. In such an example, the transistor **246** of FIGS. **2, 3,** and **4** charges the set of second capacitors **252, 256** using a current that is proportional to the current through the transistor **234** of FIGS. **2, 3,** and **4**. In example operations, the second switching signal closes the switches **260, 264** to set the switching voltage as the voltage between the capacitor **250** and the switch **244**. For example, during the operations of Block **740**, the controller circuitry **165, 232** structures the gate driver circuitry **210** to the first configuration that is illustrated in FIG. **3**.

[0117] The charging circuitry **150, 226** determines a reference voltage using the first capacitors. (Block **750**). In some examples, the controller circuitry **165, 232** disconnects the charging circuitry **150, 226** from the set of first capacitors **250, 254** responsive to the first switching signal opening the switch **244**. In such an example, the switch **244** prevents the transistor **242** of FIGS. **2, 3,** and **4** from charging the set of first capacitors **250, 254**. In example operation, the first switching signal opens the switches **258, 262** to set the reference voltage as a voltage between the capacitors **252, 256**. For example, during the operations of Block **750**, the controller circuitry **165, 232** structures the gate driver circuitry **210** to the first configuration that is illustrated in FIG. **3**. Control proceeds to Block **780**.

[0118] Although in the example of FIG. **7** operations of Block **740** occur before operations of Block **750**, the operations of Blocks **740, 750** may occur at approximately the same time. For example, the charging circuitry **150, 226** discharges the first capacitors and charges the second capacitors of the capacitor circuitry **155, 228** at approximately the same time.

[0119] If the controller circuitry **165, 232** the switching cycle is not an even switching cycle (e.g., Block **730** returns a result of NO), the charging circuitry **150, 220** charges the first capacitors to generate the switching voltage. (Block **760**). In some examples, the controller circuitry **165, 232** causes the charging circuitry **150, 226** to charge the set of first capacitors **250, 254** responsive to the first switching signal closing the switch **244**. In such an example, the transistor **242** charges the set of first capacitors **250, 254** using a current that is proportional to the current through the transistor **234**. In example operation, the first switching signal also closes the switches **258, 262** to set the switching voltage as the voltage between the capacitor **252** and the switch **248**. For example, during the operations of Block **760**, the controller circuitry **165, 232** structures the gate driver circuitry **210** to the second configuration that is illustrated in FIG. **4**.

[0120] The charging circuitry **150, 226** determines the reference voltage using the first capacitors. (Block **770**). In some examples, the controller circuitry **165, 232** disconnects the charging circuitry **150, 226** from the set of second capacitors **252, 256** responsive to the second switching signal opening the switch **248**. In such an example, the switch **248** prevents the transistor **246** from charging the set of second capacitors **252, 256**. In example operation, the second switching signal opens the switches **260, 264** to set the reference voltage as a voltage between the capacitors **250, 254**. For example, during the operations of Block **770**, the controller circuitry **165, 232** structures the gate driver circuitry **210** to the second configuration that is illustrated in FIG. **4**. Control proceeds to Block **780**.

[0121] Although in the example of FIG. **7** operations of Block **760** occur before operations of Block **770**, the operations of Blocks **760, 770** may occur at approximately the same time. For example, the charging circuitry **150, 220** charges the first capacitors and discharges the second capacitors of the capacitor circuitry **155, 228** at approximately, preferably exactly, the same time.

[0122] The comparator circuitry **170, 233** determines if the switching voltage is greater than the reference voltage. (Block **780**). In some examples, the multiplexer circuitry **160, 230** of FIGS. **1** and **2** supply the reference voltage and the switching voltage to the comparator circuitry **170, 233**. In such examples, the comparator circuitry **170, 233** compares the switching voltage and the reference voltage to control the current source circuitry **135, 220**.

[0123] If comparator circuitry **170, 233** determines that the switching voltage is not greater than the reference voltage (e.g., Block **780** returns a result of NO), control proceeds to return to Block **780**. In some examples, the comparator circuitry **170, 233** keeps the current source circuitry **135, 220** enabled (e.g., supplying current) until the switching voltage is greater than the reference voltage. In such example operations, an output of the comparator circuitry **170, 233** changes once the switching voltage is greater than or equal to the reference voltage.

[0124] If comparator circuitry **170, 233** determines that the switching voltage is greater than the reference voltage (e.g., Block **780** returns a result of YES), the comparator circuitry **170, 233** reduces the gate driver current. (Block **790**). In some examples, the comparator circuitry **170, 233** disables the current source circuitry **135, 220** responsive to a determination that the switching voltage is greater than the reference voltage. Advantageously, the comparator circuitry **170, 233** decreases the gate driver current by disabling the current source circuitry **135, 220**.

Advantageously, the comparator circuitry **170, 233** decreases the slew rate of the transistor **216** by decreasing the gate driver current. Advantageously, during a first duration of time, the transistor **216** has a relatively high slew rate, which has a relatively high-power efficiency, and during a second duration of time, the transistor **216** has a lower slew rate, which prevents excessive stress on the motor **110**. Advantageously, the gate driver circuitry **125, 210** increases power efficiency by using multiple slew rates to control the transistor **216**.

[0125] Although example methods are described with reference to the flowchart illustrated in FIG. **7**, many other methods of implementing the gate driver circuitry **125, 210** of FIGS. **1, 2, 3, 4**, and **5** may also be used in this description. For example, the order of execution of the blocks may be changed, or some of the blocks described may be changed, eliminated, or combined. Similarly,

additional operations may be included in the manufacturing process before, in between, or after the blocks shown in the illustrated examples.

[0126] FIG. 8 is a timing diagram 800 of an example switching operation of the gate driver circuitry 125, 210 of FIGS. 1, 2, 3, and 4. In the example of FIG. 8, the timing diagram 800 illustrates example drain voltages 810. The drain voltages 810 represent voltages across the transistor 216 of FIGS. 2, 3, and 4 across switching events. During the switching event of FIG. 8, the gate driver circuitry 125, 210 turns on the transistor 216, which couples the motor 110 of FIG. 1 or the load 204 of FIGS. 2, 3, and 4 to the common terminal. At a first time 820, external circuitry supplies a low-side control signal to the gate driver circuitry 125, 210 to turn on the transistor 216. At the first time 820, the voltage at the drain of the transistor 216 is a first voltage 830, which is approximately four-hundred volts (V). In the example of FIG. 8, the first voltage 830 represents the first supply voltage, which is provided at the first supply terminal.

[0127] Between the first time 820 and a second time 840, the current source circuitry 130, 135, 218, 220 of FIGS. 1, 2, 3, and 4 both supply current to generate the gate drive current. Between the first time 820 and the second time 840, the gate drive current has a first magnitude and sets the slew rate of the transistor 216 to a relatively high value. Advantageously, between the first time 820 and the second time 840, the transistor 216 has a relatively high-power efficiency responsive to the relatively high slew rate.

[0128] Between the second time 840 and a third time 850, the drain voltages 810 are approximately equal to a second voltage 860, which is approximately two-hundred volts. The second voltage 860 is a voltage approximately equal to one-half of the first voltage 830. Between the second time 840 and the third time 850, the comparator circuitry 170, 233 of FIGS. 1, 2, 3, and 4 determines that the switching voltage is greater than the reference voltage and disables the current source circuitry 135, 220. Between the second time 840 and the third time 850, the comparator circuitry 170, 233 decreases the gate drive current and the slew rate by disabling the current source circuitry 135, 220. Advantageously, decreasing the slew rate of the transistor 216 decreases the amount of stress applied to the motor 110 or the load 204 during switching operations. Advantageously, the gate driver circuitry 125, 210 increases the power efficiency of switching operations by having the relatively high slew rate for at least a portion of the switching operations of FIG. 8.

[0129] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and things, the phrase “at least one of A and B” refers to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and things, the phrase “at least one of A or B” refers to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” refers to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” refers to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0130] As used herein, singular references (e.g., “a,” “an,” “first,” “second,” etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Also, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is at least one of not feasible or advantageous.

[0131] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0132] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0133] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by at least one of the connection reference or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0134] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, or ordering in any way, but are merely used as at least one of labels or arbitrary names to distinguish elements for ease of understanding the described examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0135] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to at least one of manufacturing tolerances or other real-world imperfections. For example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified herein.

[0136] As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time+1 second.

[0137] As used herein, the phrase “in communication,” including variations thereof, encompasses one of or a combination of direct communication or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication or constant communication, but rather also includes selective communication at least one of periodic intervals, scheduled intervals, aperiodic intervals, or one-time events.

[0138] As used herein, “programmable circuitry” is defined to include at least one of (i) one or more special purpose electrical circuits (e.g., an application specific circuit (ASIC)) structured to

perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), or (ii) one or more general purpose semiconductor-based electrical circuits programmable with instructions to perform one or more specific functions(s) or operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of programmable circuitry include programmable microprocessors such as Central Processor Units (CPUs) that may execute first instructions to perform one or more operations or functions, Field Programmable Gate Arrays (FPGAs) that may be programmed with second instructions to at least one of configure or structure the FPGAs to instantiate one or more operations or functions corresponding to the first instructions, Graphics Processor Units (GPUs) that may execute first instructions to perform one or more operations or functions, Digital Signal Processors (DSPs) that may execute first instructions to perform one or more operations or functions, XPU, Network Processing Units (NPU) one or more microcontrollers that may execute first instructions to perform one or more operations or functions or integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of programmable circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more NPUs, one or more DSPs, etc., and any combination(s) thereof), and orchestration technology (e.g., application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of programmable circuitry is/are suited and available to perform the computing task(s).

[0139] As used herein integrated circuit/circuitry is defined as one or more semiconductor packages containing one or more circuit elements such as transistors, capacitors, inductors, resistors, current paths, diodes, etc. For example, an integrated circuit may be implemented as one or more of an ASIC, an FPGA, a chip, a microchip, programmable circuitry, a semiconductor substrate coupling multiple circuit elements, a system on chip (SoC), etc.

[0140] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0141] A device that is “configured to” perform a task or function may be configured (e.g., at least one of programmed or hardwired) at a time of manufacturing by a manufacturer to at least one of perform the function or be configurable (or re-configurable) by a user after manufacturing to perform the function/or other additional or alternative functions. The configuring may be through at least one of firmware or software programming of the device, through at least one of a construction or layout of hardware components and interconnections of the device, or a combination thereof.

[0142] As used herein, the terms “terminal,” “node,” “interconnection,” “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0143] In the description and claims, described “circuitry” may include one or more circuits. A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as one of or a combination of resistors, capacitors, or inductors), or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., at least one of a semiconductor die or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements or

the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by at least one of an end-user or a third-party.

[0144] Circuits described herein are reconfigurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in at least one of series or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor. While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are at least one of: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; or (iv) incorporated in/on the same printed circuit board.

[0145] Uses of the phrase “ground” in the foregoing description include at least one of a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, or any other form of ground connection applicable to, or suitable for, the teachings of this description.

[0146] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. An apparatus comprising: current source circuitry having a control terminal; charging circuitry having a first terminal and a second terminal; capacitor circuitry having a first terminal, a second terminal, a third terminal, and a fourth terminal, the first terminal of the capacitor circuitry coupled to the second terminal of the charging circuitry, the second terminal of the capacitor circuitry coupled to the third terminal of the charging circuitry; and comparator circuitry having a first terminal, a second terminal, and an output terminal, the first terminal of the comparator circuitry coupled to the third terminal of the capacitor circuitry, the second terminal of the comparator circuitry coupled to the fourth terminal of the capacitor circuitry, the output terminal of the comparator circuitry coupled to the control terminal of the current source circuitry.
2. The apparatus of claim 1, wherein the charging circuitry includes: a first transistor having a first terminal and a control terminal; a first switch having a first terminal and a second terminal, the first terminal of the first switch is coupled to the first terminal of the first transistor, the second terminal of the first switch coupled to the first terminal of the capacitor circuitry; a second transistor having a first terminal and a control terminal, the control terminal of the second transistor is coupled to the control terminal of the first transistor; and a second switch having a first terminal and a second terminal, the first terminal of the second switch is coupled to the first terminal of the second transistor, the second terminal of the second switch is coupled to the second terminal of the capacitor circuitry.
3. The apparatus of claim 1, wherein the capacitor circuitry includes: a first capacitor having a first terminal and a second terminal, the first terminal of the first capacitor coupled to the first terminal of the charging circuitry; a second capacitor having a terminal coupled to the first terminal of the comparator circuitry and the second terminal of the first capacitor; and a third capacitor having a

terminal coupled to the second terminal of the charging circuitry and the second terminal of the comparator circuitry.

4. The apparatus of claim 1, wherein the capacitor circuitry includes: a first capacitor having a terminal coupled to the second terminal of the charging circuitry and the first terminal of the comparator circuitry; a second capacitor having a first terminal and a second terminal, the first terminal of the second capacitor coupled to the third terminal of the charging circuitry; and a third capacitor having a terminal coupled to the second terminal of the comparator circuitry and the second terminal of the second capacitor.

5. The apparatus of claim 1, wherein the capacitor circuitry further has a fifth terminal and a sixth terminal, the apparatus further comprising multiplexer circuitry having a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, and a sixth terminal, the first terminal of the multiplexer circuitry is coupled to the third terminal of the capacitor circuitry, the second terminal of the multiplexer circuitry is coupled to the fourth terminal of the capacitor circuitry, the third terminal of the multiplexer circuitry is coupled to the fifth terminal of the capacitor circuitry, the fourth terminal of the multiplexer circuitry is coupled to the sixth terminal of the capacitor circuitry, the fifth terminal of the multiplexer circuitry is coupled to the first terminal of the comparator circuitry, the sixth terminal of the multiplexer circuitry is coupled to the second terminal of the comparator circuitry.

6. The apparatus of claim 1, wherein the current source circuitry further has an output terminal, the charging circuitry further has a third terminal, the apparatus further comprising: a capacitor having a first terminal and a second terminal, the first terminal of the capacitor is coupled to the third terminal of the charging circuitry; and a transistor having a first terminal and a control terminal, the first terminal of the transistor coupled to the second terminal of the capacitor, the control terminal of the transistor coupled to the output terminal of the current source circuitry.

7. The apparatus of claim 6, further comprising current multiplication circuitry having a first terminal and a second terminal, the first terminal of the current multiplication circuitry coupled to the first terminal of the capacitor and the third terminal of the charging circuitry, the second terminal of the current multiplication circuitry coupled to the output terminal of the current source circuitry, and the control terminal of the transistor.

8. The apparatus of claim 1, wherein the current source circuitry is first current source circuitry, the first current source circuitry further has an output terminal, the apparatus further comprising second current source circuitry having an output terminal coupled to the output terminal of the first current source circuitry.

9. An apparatus comprising: capacitor circuitry including: a set of first capacitors; and a set of second capacitors; charging circuitry coupled to the capacitor circuitry, the charging circuitry configured to: determine a reference voltage using the set of first capacitors; and generate a switching voltage by charging the set of second capacitors; and comparator circuitry coupled to the capacitor circuitry, the comparator circuitry configured to decrease a drive current responsive to a comparison of the reference voltage to the switching voltage.

10. The apparatus of claim 9, further comprising controller circuitry coupled to the charging circuitry, the controller circuitry configured to: generate a first control signal by dividing a pulse width modulation (PWM) signal; generate a second control signal by inverting the first control signal; cause the charging circuitry to generate the reference voltage responsive to the first control signal; and cause the charging circuitry to generate the switching voltage responsive to the second control signal.

11. The apparatus of claim 9, wherein the charging circuitry is further configured to: generate the switching voltage by charging the set of first capacitors; and determine the reference voltage using the set of second capacitors.

12. The apparatus of claim 11, further comprising multiplexer circuitry coupled between the capacitor circuitry and the comparator circuitry, the multiplexer circuitry configured to: supply the

switching voltage from one of the set of first capacitors or the set of second capacitors to the comparator circuitry; and supply the reference voltage from one of the set of first capacitors or the set of second capacitors to the comparator circuitry.

13. The apparatus of claim 9, further comprising current source circuitry coupled to the comparator circuitry, the current source circuitry configured to supply the drive current based on the comparison of the reference voltage to the switching voltage.

14. The apparatus of claim 13, wherein the current source circuitry is first current source circuitry, the apparatus further comprising second current source circuitry coupled to the first current source circuitry, the second current source circuitry configured to increase the drive current.

15. The apparatus of claim 13, further comprising a transistor coupled to the current source circuitry, the transistor configured to: decrease a voltage of the transistor at a first slew rate responsive to a first current from the current source circuitry; and decrease the voltage of the transistor at a second slew rate responsive to a second current from the current source circuitry, the second current is less than the first current.

16. An apparatus comprising: a transistor having a control terminal; and current source circuitry having a first terminal and a control terminal, the first terminal of the current source circuitry coupled to the control terminal of the transistor; comparator circuitry having a first terminal, a second terminal, and an output terminal, the output terminal of the comparator circuitry coupled to the control terminal of the current source circuitry; and capacitor circuitry having a first terminal and a second terminal, the first terminal of the capacitor circuitry coupled to the first terminal of the comparator circuitry, the second terminal of the capacitor circuitry coupled to the second terminal of the comparator circuitry.

17. The apparatus of claim 16, wherein the capacitor circuitry further has a third and fourth terminal, the apparatus further comprising charging circuitry having a first terminal and a second terminal, the first terminal of the charging circuitry is coupled to the third terminal of the capacitor circuitry, and the second terminal of the charging circuitry is coupled to the fourth terminal of the capacitor circuitry.

18. The apparatus of claim 17, wherein the transistor is a first transistor, the charging circuitry includes: a second transistor having a first terminal and a control terminal; a first switch having a first terminal and a second terminal, the first terminal of the first switch is coupled to the first terminal of the second transistor, the second terminal of the first switch coupled to the third terminal of the capacitor circuitry; a third transistor having a first terminal and a control terminal, the control terminal of the third transistor is coupled to the control terminal of the second transistor; and a second switch having a first terminal and a second terminal, the first terminal of the second switch is coupled to the first terminal of the third transistor, the second terminal of the second switch is coupled to the fourth terminal of the capacitor circuitry.

19. The apparatus of claim 16, wherein the capacitor circuitry includes: a first capacitor having a terminal; a second capacitor having a terminal coupled to the first terminal of the comparator circuitry and the second terminal of the first capacitor; and a third capacitor having a terminal coupled to the second terminal of the comparator circuitry.

20. The apparatus of claim 16, wherein the capacitor circuitry further has a third terminal and a fourth terminal, the apparatus further comprising multiplexer circuitry having a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, and a sixth terminal, the first terminal of the multiplexer circuitry is coupled to the first terminal of the capacitor circuitry, the second terminal of the multiplexer circuitry is coupled to the second terminal of the capacitor circuitry, the third terminal of the multiplexer circuitry is coupled to the third terminal of the capacitor circuitry, the fourth terminal of the multiplexer circuitry is coupled to the fourth terminal of the capacitor circuitry, the fifth terminal of the multiplexer circuitry is coupled to the first terminal of the comparator circuitry, the sixth terminal of the multiplexer circuitry is coupled to the second terminal of the comparator circuitry.

